

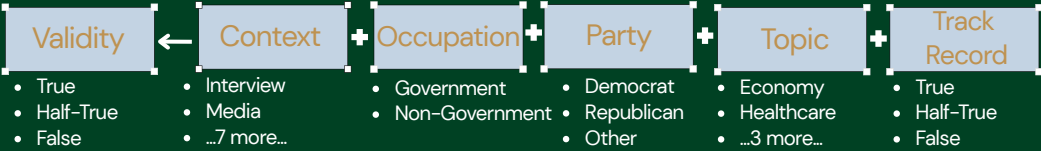
Introduction

Our data consists of over 1300 political statements organized by [Politifact.com](#) and gathered from unique speeches, news reports, and more recorded between 2007 –2017. Each statement is classified in a 5 tier spectrum of validity, from false to true. Before we begin analyses, we must aknowledge that this data comes from news outlets, thus we expect statement inccuracy and subjectivity considering media often skews information for viewership. This can be extended into a cultural context, where each news outlet approaches conventions differently, based on the characteristics of their viewer base. We also understand that historically, viewership from more privelideged backgrounds have had more access to these sources. As we navigate this data, we seek to understand

- 1.What factors are associated with the validity of a political statement
- 2.What linguistic features are associated with the validity of a political statement

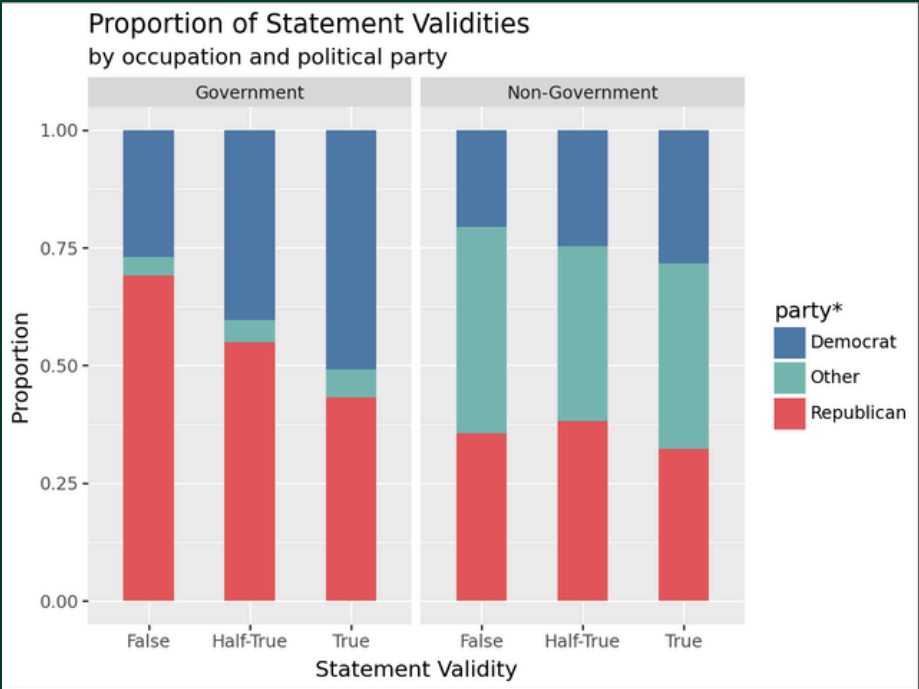
Methods

- As a result of unorganized raw data:
- 1.Created functions to organize messy qualitative variables into levels
 - 2.Joined and standardized quantitative variables for accurate modeling and interpretation
- To study the data:
1. Fit a multinomial logistic regression model to predict a political statement's validity (true, half-true, false)
 - 2.Implemented Grid Search to optimize the parameters and performance of the model
- The model considered the context of the statement, the topic, the speaker's occupation and political party, and the track record of the speaker in each of the 3 statment validity categories.



- To explore lingusitic features involved in the validity of a statement:
- 1, Implemented Tf-IDF Vectorizer. We feature engineerined features involving a statments punctuation and a statements words part-of-speech., using library nltk.
 - 2.Fit the logistic regression model onto data to pull out the coefficients.
- Allows us to see the most important words for both true and fake news, indicated magnitude and sign of the coefficients.

Visualizing Preliminary Data



- This chart visualizes the preliminary data (Validities, Occupation, Party):
1. Statement proportions for democrats ascend as statements get more true
 2. The opposite effect for republicans, across both occupation types.

Ethics/Limitations

With respect to the nature of the dataset, our results can be applied to political statements from 2007–2017 in the United States.

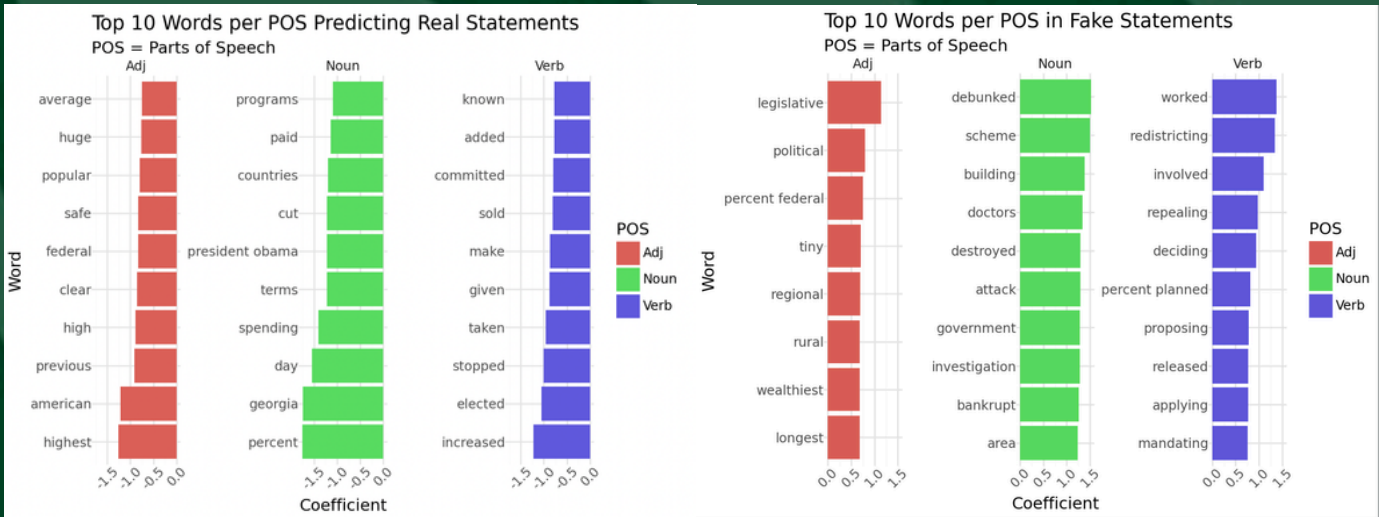
- Potential Ethical and Model Limiations:
- 1.Reducing statement validity from 5 to 3 levels may be blurring the lines b/w what statements are really ‘purely true’ and ‘purely false’.
 - 2.TF-IDF limited classification labels of validity into 2 levels as well.
 3. The data was manually classified, introduces subjectivity in what is classified as true and false.
 4. Imbalanced data b/w false statements compared to other 2 levels, which redcuded the strength of our models prediction
 5. Association of words with the validity of a statement does not imply causation, as a result of unknown sampling methods.
 - 6.NLTK library used for TF-IDF misclassified some verbs as nouns.

Results

Cross-Val F1-Score		Cross-Val Accuracy
0.437		0.418
Statement Validity	Political Party	Track Record
FALSE	1.23 (Republican)	1.42 (False TR)
HALF TRUE	0.989 (Republican)	2.46 (H-T TR)
TRUE	0.8 (Republican)	2.41 (True TR)

Table of odds for explanatory variables Track Record with categories (True, Half True, False) and Politcal Party (Democrat, Republican, Other). The baseline for Political Party is Democrat. Response variable is statement validity (True, Half True, False).

The strongest associations being among track records and their corresponding validity classes. “For each 1 std. dev. increase in the statement’s speaker’s truthful track record, the odds of that statment being classified as true is multiplied by a factor of 2.41. ” Political party also contained stronger associations. Compared to Democrats, republican statements have higher odds of being classified as False by 23%. Other less notable variables had weaker associations with the validity of a statment.



Graphs showing the top 10 words of each group of POS (whether it's a noun, adjective or verb). The left is for statements that turned out to be true and the right is for statements that turned out to be false. Statement validity was shortend to 2 groups (false and true).

After using a TfIdVectorizer with ngram_range set to 1–2 words, we grouped by our “fake” column. Real and fake classified news have similar word counts per statement as well, with fake being slightly higher. Then, we plotted the top 10 most frequent adjectives, verbs, and nouns for fake and true news from the POS engineering. Despite some incorrect classification of nouns by the POS, we found it interesting that the most common words associated with fake news had more of a negative/violent connotation than the true news, which was more professional.

